

Closing the Research Loop

Decay-Driven Dynamic Re-Mining:
Design, Evidence, and the Experiment That Tests It

28 September 2026

Abstract

The system mines alpha factors once, from a human-written direction, and trades the result indefinitely. On the first genuinely leak-free CSI300 window this does not survive: rank IC decays from 0.1104 to ~ 0.045 within three years and cumulative excess return over the cap-weighted, dividend-reinvested CSI300 is -41.3% across 2017–2025. We present (i) the one intervention already measured to work — an annual combiner refit, which turns a -18.8% cumulative excess into $+2.7\%$ on *unchanged* factors; (ii) a design that closes the research loop by monitoring decay and re-mining against it, with deterministic rules as a hard floor and an LLM agent permitted only to accelerate them; and (iii) a nine-arm historical replay that isolates every mechanism, including two non-LLM control arms that test for temporal leakage at a layer the main ablation cannot reach. We then argue, from metrics rather than assertion, that this work closes the *research* half of the quant loop and cannot close the *trading* half: predictive power is present and industry-normal, and the shortfall is located entirely in portfolio construction.

1 Where we are

All figures below use protocol `protocol.csi300_lightgbm_topk_decay.yaml`: fit 2005-01-01 to 2016-12-31 (purged to 2016-12-09), scored 2017-01-01 to 2025-12-31, LightGBM combiner, `topk_dropout` construction ($k = 50$, `n.drop`= 5, `max_weight`= 0.03), the full κ cost model, and benchmark SH000300TR — the official cap-weighted CSI300 *total-return* index. Because `costs.net_return` computes gross – benchmark – charges, every return reported here is **excess**, and CRR denotes **cumulative excess return**.

Table 1: Leak-free out-of-sample baseline, 2017–2025.

configuration	IC _{rank}	ARR	CRR	IR	TC
MAIN-FULL (147 factors)	0.0565	-5.96%	-41.32%	-0.32	0.37
ORIGINAL (147 factors)	0.0488	-9.78%	-59.07%	-0.59	0.33
null model (no factors)	—	-14.33%	-73.87%	-1.05	—

Two properties of this baseline drive everything that follows.

Signal decays on a roughly three-year clock. Rank IC by year for MAIN-FULL: 0.1104, 0.0898, 0.0587, 0.0543, 0.0504, 0.0524, 0.0160, 0.0288, 0.0469. The decline is monotone in distance from the fit window and then plateaus near 0.045. The 2023 trough is genuine: IC goes negative for the reduced zoo that year.

The returns are beta, not alpha. Correlation of annual excess return with the *benchmark's* return is **−0.73**; with the book's own rank IC it is +0.11. Signal quality explains about 1% of the variance in excess return. Every profitable year is a year the index fell. The decisive case is 2017: rank IC 0.1104, the highest this system has ever produced out of sample and above every break-even estimate, returned −5.45% because the index rose 24.25%.

2 What already works: the annual combiner refit

The cheapest available intervention changes no factors at all. It refits the LightGBM combiner each year on data strictly before the year being traded (fit $\leq Y - 1$, report Y , expanding window). The factor library is frozen throughout, so any gain is attributable to the refit alone.

Table 2: MAIN-FULL, 147 factors: frozen 2016 fit versus annual rolling refit. Both columns are excess over SH000300TR, net of κ costs.

year	frozen (fit ≤ 2016)		annual refit		improvement	
	IC _{rank}	ARR	IC _{rank}	ARR	Δ IC _{rank}	Δ ARR
2021	0.0504	+5.82%	0.0586	+11.40%	+0.0082	+5.58 pp
2022	0.0524	+1.81%	0.0598	+8.98%	+0.0074	+7.17 pp
2023	0.0160	−7.42%	0.0298	−1.99%	+0.0138	+5.43 pp
2024	0.0288	−4.31%	0.0533	−3.75%	+0.0245	+0.56 pp
2025	0.0469	−14.94%	0.0616	−10.25%	+0.0147	+4.69 pp
mean	0.0389	—	0.0526	—	+35%	+4.69 pp
stitched		−4.24%		+0.55%		
CRR		−18.8%		+2.68%		+21.5 pp

Every one of the five years improves, on both rank IC and ARR. Mean rank IC rises 35%; the transfer coefficient rises from 0.366 to 0.386. **This is the only configuration of the nine we have tested with a positive CRR**, and it is currently **not done at all** — the production system fits once and never refits. The frozen stitched figure is compounded from the per-year values; the refit figure is computed on the concatenated daily series.

Note also that 2023 — the year attributed in the literature to an A-share regime rotation — more than halves its loss under a refit, and its rank IC rises 86%. Roughly half of what reads as a regime break is combiner staleness.

Factor freshness is worth about twice as much again. Fresh factors score 0.1001 (2017–18) and 0.1136 (2016) against 0.0526 for stale factors with a fresh combiner and 0.0439 for both stale. Combining the two through $IR = TC \cdot IC \cdot \sqrt{N}$: both stale 0.30, refit-only 0.36, re-mined and refit **0.68** — the first configuration above the conventional 0.5 viability bar. *This figure carries a caveat that the experiment is designed to test; see §4.1.*

3 The design

Mining cadence is the primary lever and combiner refit the secondary one, and both belong inside one loop. The design has a deterministic layer that always runs and an advisory layer that may only make it act sooner.

Deterministic layer (the hard floor). Per-factor rolling 63-day IC is compared against the factor’s validation IC at admission. Tiers, on *sustained* breaches rather than touches: *healthy* ($\geq 70\%$ of baseline); *soft decay* ($< 70\%$ for 30+ days) $\rightarrow$ half weight; *hard decay* ($< 50\%$ for 60+ days) $\rightarrow$ removed from scoring, UUID retained and re-tested quarterly. Cadences: monitor **daily**; refit **monthly** floor; re-mine **annual** floor, fired earlier when $\geq 33\%$ of the active set is hard-decayed or the book’s own rolling IC breaches 50% of baseline for 60 days.

Annual is the *minimum* defensible re-mine cadence, not a cautious one: the published estimate for a medium-frequency factor’s half-life is now around 18 months, against 5–7 years pre-AI.

Advisory layer (the decay agent). Because the deterministic rule cannot fire before a 30-day sustained breach, there is a structural 30–60 day latency to buy back. An LLM agent runs weekly and on every tier transition, and may: trigger a refit or re-mine early, or demote a factor early. It may delay exactly one thing — a *soft* demotion, via a regime exemption that must cite a computed market condition, expires after one quarter, may never cover more than 25% of the active set, and may never apply to hard decay. It may never postpone a scheduled refit or re-mine. A malformed, timed-out, or absent agent is a non-event: the deterministic layer proceeds unchanged.

Soft decay under a tree model. “Half weight” is not literally implementable: gradient-boosted trees are invariant to monotone feature scaling and the LightGBM path produces no per-factor weights. The combiner already fits one model per seed and averages the predictions, so soft decay is expressed as **feature-subset bagging**: the factor’s column enters only $\lceil p \cdot K \rceil$ of the K seed models, with $p \in \{1, 0.5, 0\}$ reproducing the tier multipliers exactly.

The library, and what actually trades. The library is append-only and unbounded: every factor ever mined, admitted or rejected, retained under its UUID with its full trajectory. What trades is a *capped, gated* active set, selected per refit by tier, then pathology, then an orthogonality gate ($\rho_{\max} \leq \bar{\rho}$, so no two bets are the same — currently disabled by default), then capacity-with-replacement, then re-measured marginal contribution against the set as it now stands.

Direction selection, without a news corpus. Each re-mine derives its own directions from a **computed market-state report** over data $\leq T$: cross-sectional dispersion, breadth, the size-decile spread, volatility term structure, return autocorrelation at 1/5/20 days, turnover, overnight-gap share, industry dispersion, and mean pairwise correlation. No news is read. This is deliberate: the 2023 rotation *is* a rise in dispersion and an inversion of the size spread, and the agent sees those as numbers with no label. Nothing about the regime narrative is ever stated to the model.

Parent selection. Selection today is pure exploitation over a pool destroyed at the end of each run, and it has produced operator monoculture twice. The library becomes a MAP-Elites style archive binned by operator class, horizon, regime profile and decay tier, with parents drawn on a logged budget: 50% exploit (best by shrunk marginal contribution among healthy factors), 30% explore (sparse archive cells), 20% resurrect (hard-decayed factors whose regime profile matches the current market state). Selection within a pool is UCB over *cells*, not factors, because individual fitness is noisy enough that three of four verdicts flip across combiner seeds.

4 The experiment

Validation is a historical replay: at each trigger point T , everything — market state, direction, mining, admission, active-set gate, combiner fit — uses data $\leq T$ only; the following year is scored and never revisited; per-arm daily returns are stitched into one continuous series.

Table 3: The nine arms. “Schedule” distinguishes the deterministic floor alone from the floor plus the advisory decay agent.

arm	directions	schedule	re-mines	what it isolates
C	—	none	no	static baseline; the number to beat (CRR -41.32%)
D	—	calendar	no	refit only; already measured (CRR $+2.68\%$)
A	fixed	deterministic	yes	the value of re-mining itself
B	agent	deterministic	yes	the direction agent (vs A)
G	fixed	+ decay agent	yes	the decay agent (vs A)
H	agent	+ decay agent	yes	the full system; interaction of both agents
H⁻	agent	+ agent, accelerate-only	yes	the regime-exemption mechanism (vs H); run only if H beats G
E	Alpha158	calendar	no	leakage control: a fixed, published, pre-LLM factor set
F	random DSL	calendar	no	leakage control: mechanically generated expressions, no survivorship

Arms A, B, G, H form a 2×2 factorial over $\{\text{fixed, agent}\}$ directions $\times$ $\{\text{deterministic, +agent}\}$ scheduling. The contrasts of interest are $A - D$ (re-mining), $B - A$ (direction agent), $G - A$ (decay agent), $H - G$ and $H - B$ (each agent in the presence of the other), and the interaction $(H - G) - (B - A)$.

Metrics. Primary: stitched **CRR**. Secondary, per year and per arm: ARR, IR, MDD, rank IC, IC, ICIR, TC, turnover, cost in bps, active-set size, admissions, and the decay tier census. The benchmark’s own return is reported alongside every arm, because excess return here is beta-driven and an arm that “wins” in a falling market has not necessarily produced alpha. Deflated Sharpe is computed against the *true cross-mine trial count*, which grows with every mine and correctly raises the bar.

Cost. Annual triggers across 2017–2025 give nine points per re-mining arm. Arms G and H share mines with A and B wherever their trigger dates coincide, so the incremental cost of the decay-agent arms is only the divergences; the estimate is 22–26 mines in total. Arms C, D, E and F require no mining at all. The per-mine wall-clock and LLM cost is measured before anything is launched, and a reduced three-point trigger set is the pre-committed fallback.

4.1 Temporal leakage, and why two control arms

The generating model’s weights contain the future; no prompt retracts that. Three layers bound what leakage can do — context hygiene (only the train window is disclosed, clamped to the day before the earliest fold’s validation day, failing closed), an output validator rejecting dates, tickers

and named events, and a specificity cap so a direction names mechanisms rather than answers. But bounds are not measurements, so the experiment measures two distinct exposures.

Late-sample leakage: already refuted. If hindsight had been exploited, the factors would hold up in the leaked period. They do not: rank IC falls monotonically with distance from the fit and troughs at 0.0160 in **2023**, the most-documented regime break in the sample and precisely where a leaking model’s knowledge is densest. There is also a structural reason — admission selects on data $\leq T$, so a factor whose edge exists only in the future is rejected before entering the library.

First-year leakage: *not* refuted, and it matters. Within the pool that works before T , hindsight can still select those the model knows persist longest. That inflates **year one** and converges to the honest baseline as even the leaked picks decay — reproducing the observed curve exactly. The 2023 trough does not exclude it, since by 2023 a leaked pick is dead too. And the A versus B ablation is **blind** to it, because both arms author factors with the same model: this leakage is common mode and differences out.

This bears directly on the project’s central justification. The +90% factor-freshness figure, which carries re-mined-and-refit to gross IR 0.68, is a *first-year* measurement — and an annually re-mining loop lives on exactly that number.

Hence arms **E** and **F**, which contain no LLM in generation and therefore cannot leak. Arm E is Alpha158, a fixed published pre-LLM set; arm F is mechanically generated DSL expressions, which arm E cannot replace because Alpha158 is a *survivor* set and may show unusually flat decay for reasons unrelated to leakage. The test compares the **ratio** $\text{IC}_{\text{rank}}^{\text{year-1}} / \text{IC}_{\text{rank}}^{\text{steady-state}}$ (a ratio, because the controls are weaker in level). A materially larger ratio for the mined arms is the leakage signature, isolated at the layer the main ablation cannot reach. **Until these arms clear year one, the deployable expectation is the steady-state rank IC ($\sim 0.045\text{--}0.05$), not the first-year value (~ 0.11), and +90% is quoted as an upper bound.**

5 What this closes, and what it cannot

5.1 The research half is present and industry-normal

Table 4: Predictive-power metrics. These are construction-, cost- and benchmark-independent: they are properties of the signal alone.

metric	full 2017–2025	2017	typical usable range
IC	0.0425	0.0941	0.02–0.05
Rank IC	0.0565	0.1104	0.03–0.05
ICIR	0.315	0.743	0.3–0.5
Rank ICIR	0.409	0.845	0.3–0.5

On leak-free out-of-sample data the mined library sits inside the normal range for a usable equity signal on every predictive-power metric, and its first out-of-sample year is comfortably above it. **The system finds alpha.** That is the research question, and it is answered affirmatively.

5.2 The trading half is where it is lost, and the metrics locate it exactly

Write the standard decomposition $IR = TC \cdot IC \cdot \sqrt{N}$. With $IC = 0.0425$, $N \approx 300$ so $\sqrt{N} = 17.3$, and $TC = 0.37$:

$$IR_{\text{gross}} = 0.37 \times 0.0425 \times 17.3 \approx 0.27\text{--}0.36.$$

Every term is individually normal. **A transfer coefficient of 0.37 is squarely inside the industry range of 0.3–0.6 for a long-only book** — there is no anomalous loss to recover. The product is sub-viable because of the *mandate*: $N \approx 300$ names, $k = 50$ holdings, long-only, benchmarked to a cap-weighted index with no hedging instrument. Five independent measurements confirm the constraint is construction, not signal:

1. **The null model reproduces the pathology.** A book holding *no* mined factors shows correlation -0.64 between excess return and the benchmark, against -0.73 for MAIN-FULL. The beta shortfall is a property of the construction, not of the factors.
2. **The position cap is not binding.** Raising `max_weight` from 0.03 to 0.10 left TC at 0.36 and cost 10.5 pp. The mean-variance optimiser *diversified* from 100 to 134 names rather than concentrating — a total-risk optimiser answering a benchmark-relative mandate.
3. **Relaxing long-only raises TC and still loses.** Signed (dollar-neutral) top- k lifted TC from 0.37 to **0.51** at an unchanged cap — confirming long-only, not the cap, is the binding constraint — yet absolute long/short return was -1.65% /yr with 2 of 9 positive years.
4. **Three of nine tested configurations are worse than holding no factors at all**, including Kelly/mean-variance at CRR -78.95% against the null model’s -73.87% .
5. **The best signal year is a losing year.** 2017, rank IC 0.1104, ICIR 0.743: -5.45% , because the index rose 24.25%.

5.3 The asymmetry, stated plainly

Table 5: What each half of the loop actually has.

Factor side (research)	Portfolio side (trading)
LLM search loop with mutation, crossover and refinement	<code>topk</code> = 50
Admission gate on marginal contribution tested against its own noise	<code>max_weight</code> = 0.03
Benjamini–Hochberg FDR control; deflated Sharpe against trial count	<code>n_drop</code> = 5
Orthogonality gate, diversity scoring, lineage and verdict tracking	
Per-factor tear sheets, diagnosis and routing	
<i>A research system.</i>	<i>Three constants, fixed once by a 54-config sweep on a single 2021 window.</i>

The objective the search maximises is a function of IC. **Nothing in the system optimises IR or CRR.** The re-mining design in this document makes the research half close properly — it monitors its own decay, re-mines against it, and refits continuously. It will raise IC, and on the measured decomposition that raises gross IR from 0.30 to perhaps 0.68.

It will not fix a transfer coefficient that is already normal, a benchmark that cannot be tracked long-only, or a construction stage with no search loop, no admission criterion and no multiple-testing control. **Converting demonstrated predictive power into realised profit requires a second system, built for construction the way this one was built for factors** — searching over constraints, hedges, weighting schemes and rebalancing rules, with an objective that is the realised net IR rather than the signal's IC.

The falsifiable form of this claim. If the conclusion is wrong, the replay will say so: an arm that materially improves IC should then also improve CRR. The measured relationship says it will not — correlation between annual excess return and rank IC is $+0.11$, against -0.73 with the benchmark. We will report whichever answer the data gives.

6 What is not tested here

Shorting and borrow costs (deferred to the construction phase, and the signed result above says it is not a quick win); live news; real capital and a broker integration; and any change to the construction stage itself. NAV is simulated and compounds from ¥100M, which at that scale leaves the market-impact term effectively inert — it matters as the capital base changes, not now.